

THE DOUBLE-EDGED SWORD OF DECENTRALIZED PLANNING IN MULTITEAM SYSTEMS

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The literature on small stand-alone teams has suggested that decentralization has predominantly positive features. However, in multiteam systems, the presence of other highly interdependent teams adds a level of complexity that may preclude generalizing from teams to multiteam systems. We studied the effects of decentralized planning in 210 multiteam systems, each composed of three six-person, functionally specialized component teams. As has previous research, we find that decentralized planning has positive effects on multiteam system performance, attributable to enhanced proactivity and aspiration levels. However, we also find that the positive effects associated with decentralized planning are offset by the even stronger negative effects attributable to excessive risk seeking and coordination failures. We discuss the implications for theories of intrateam and interteam dynamics, along with the applied implications for designing empowerment interventions in team and multiteam contexts.

Many organizations strive to remain competitive in markets that are increasingly volatile, competitive, and dynamic. To adapt to changing environments, many organizations have shifted from individual-based work toward team-based structures (Kozlowski & Bell, 2003; Lawler, Mohrman, & Ledford, 1995). Teams enable organizations to be adaptive, because they are dynamic and flexible systems comprised of individuals who possess diverse skills and experiences (Hoffman & Maier, 1961;

McGrath, Arrow, & Berdahl, 2000). This shift in the nature of organizational structures has, in turn, led to a growing interest in determinants of team effectiveness (LePine, Piccolo, Jackson, Mathieu, & Saul, 2008). Much of the research on team effectiveness, however, has studied team processes within stand-alone teams, despite the well-acknowledged fact that work teams are typically integrated into larger organizational units (Ilgen, Hollenbeck, Johnson, & Jundt, 2005).

As a result, even though interteam dynamics play an important role in team-based structures, there is a notable imbalance between studies focused on intrateam processes relative to studies on interteam processes (Alderfer & Smith, 1982; Ancona & Caldwell, 1992). Research addressing this imbalance is critical, because many of the factors that contribute to effective processes *within* teams may create difficulties in coordination *between* teams. For example, strong intragroup cohesiveness along with feelings of potency and personal identification within groups can create competition and conflict between groups (Tajfel, 1982). In addition, effective within group processes often aim to pro-

We gratefully acknowledge the generous support provided by the faculty, staff, and students of the U.S. Air Force Squadron (USAF) Officers College, Maxwell Air Force Base, Montgomery, Alabama. In particular, we thank Gen. Donald Cook, USAF, Retired; Col. Michael Tanous, USAF; Maj. Caroline Knutson-Vacarro, USAF, Retired; Maj. Kenneth Hanson, USAF, Retired; and Dr. Fil Arenas and Dr. Matthew Stafford. Special thanks go to Dr. Hank Dasinger for his vision and leadership in managing the multiteam system that was required to accomplish this research. We are also thankful for the help provided by John Karasinski and Benjamin Thomas Jacquier during the data-coding process.

mote role expansion and flexibility, which enhance the effectiveness of small stand-alone teams. However, in tightly coupled systems of teams, flexibility on the part of one team can become unpredictability for other teams that need to coordinate their efforts with the focal team. The relative imbalance between the amount of research on within-team versus between-team behavior has led to many calls for more research focused on “teams of teams” (Hoegl, Weinkauff, & Gemuenden, 2004) or multi-team systems (Mathieu, Marks, & Zaccaro, 2001).

Multiteam systems are unique organizational arrangements composed of two or more teams (hereafter referred to as “component teams”), which share input, process, and outcome interdependencies as they pursue a common goal (Mathieu et al., 2001; Zaccaro, Marks, & DeChurch, 2012). The success of multiteam systems is tightly dependent on both within- and between-team processes (Marks, DeChurch, Mathieu, Panzer, & Alonso, 2005). However, unlike in small stand-alone teams, the large size of multiteam systems, along with the specialized nature of different members’ skills, makes direct, face-to-face, real-time communication among all team members difficult, if not impossible. In addition, due to inherent interdependencies and tight coupling associated with multiteam systems, failure on the part of any one component team usually leads to failure at the whole multiteam system level. Illustrating this point, the relief multiteam system that was formed by the Red Cross and Federal Emergency Management Agency (FEMA) to help Hurricane Katrina victims was criticized for not meeting its goals efficiently. Governmental auditing reports noted that the Red Cross was unclear about the coordinating role and responsibilities of FEMA. Both teams spent time trying to establish operations and procedures, rather than focusing on services (U.S. Government Accountability Office, 2006).

To advance theory and research on teams and multiteam systems, it is critical to test the degree to which well-accepted theories derived for stand-alone teams generalize to multiteam systems, given the potential boundary conditions associated with larger size, higher specialization, and tighter interdependence between teams in such systems (Davison, Hollenbeck, Barnes, Slesman, & Ilgen, 2012). Thus, multiteam systems may present an important boundary condition to widely accepted theories of small teams. We explore this possibility for decentralization, which is generally recognized as a positive design feature for small stand-alone teams

(e.g., Boone & Hendriks, 2009; Hirst, Van Knippenberg, Chen, & Sacramento, 2011). We take a general approach to multiteam systems and propose that, unlike in small stand-alone teams, decentralization does not have only positive outcomes in multiteam systems. We integrate theories of decentralization, goal setting, and proactivity and examine theoretically and empirically the consequences of decentralized planning in a multiteam system context.

Decentralized planning gives lower-level teams and individuals within organizations a voice in tactical directions and operations of their units (Mintzberg, 1979). It provides lower level units with more power and control over important decisions and puts leadership in an advisory role (Burke, 1986; Kanter, 1993; Mintzberg, 1979, 1983; Tannenbaum, 1968). The study of decentralized planning in multiteam systems has important practical implications, because decentralization is regarded primarily as a positive feature for small stand-alone teams (e.g., Boone & Hendriks, 2009; Hirst et al., 2011; Ling, Simsek, Lubatkin, Lyon, & Veiga, 2008; Mehra, Smith, Dixon, & Robertson, 2006) and serves as a core design element in empowerment interventions (e.g., Aryee, Walumbwa, Seidu, & Otaye, 2012; Bartunek & Spreitzer, 2006; Seibert, Wang, & Courtright, 2011; Spreitzer & Doneson, 2007). In addition, literature on shared leadership in small teams suggests that decentralized structures that promote participative decision making benefit overall team performance (e.g., Carson, Tesluk, & Marrone, 2007; Pearce & Sims, 2002).

Whereas the literature on stand-alone teams supports the use of decentralizing interventions that shift responsibility and authority from leadership to team members (e.g., Chen, Sharma, Edinger, Shapiro, & Farh, 2011; Cordery, Morrison, Wright, & Wall, 2010; DeJoy, Wilson, Vandenberg, McGrath-Higgins, & Griffin-Blake, 2010; Gibson, Porath, Benson, & Lawler, 2007), the literature on large-scale organizations suggests that there are both virtues and liabilities to decentralized structures. Moreover, because multiteam systems lie at the nexus between small teams and large organizations, we may not be able to generalize the prescriptions associated with decentralizing interventions in stand-alone teams to larger and more specialized systems of teams (Drazin & Van de Ven, 1985; Thompson, 1967). Indeed, the possibility that the positive effects of decentralized planning at the component team level might be more than offset at the higher multiteam system level, while plausible,

has never been tested. Our current work addresses this possibility and argues that although there are likely to be positive influences of decentralized planning on multiteam system performance attributable to increased proactivity and aspiration levels, there are also likely to be negative influences attributable to excessive risk seeking and coordination problems that might negate or even reverse the traditional effects.

This study contributes to literature and theory on teams and multiteam systems in three important ways. First, we argue theoretically and then show empirically that decentralization, which is viewed as a predominantly beneficial practice in the small teams' literature, may actually harm performance in multiteam systems, thus establishing an important boundary condition for well-accepted theories in the teams' literature (e.g., empowerment, autonomous work designs, self-management). Second, we identify the mediating mechanisms through which decentralized planning is likely to benefit as well as to harm multiteam system performance. Third, we utilize a situated experiment to test these ideas in a naturalistic environment that combines the beneficial elements of both laboratory and field research.

THEORIES OF DECENTRALIZATION AND GOAL SETTING: AN INTEGRATIVE APPROACH TO MULTITEAM SYSTEMS

Multiteam systems are becoming increasingly popular in organizational contexts, because they are large enough to address very complex problems but flexible enough to respond to problems or opportunities in real time (Zaccaro et al., 2012). Multiteam systems may or may not cross organizational boundaries but, in all cases, consist of highly interdependent component teams that typically come together to respond to a unique organizational need (Mathieu et al., 2001). They tend to be larger and more specialized than traditional stand-alone teams but smaller and less formalized than large-scale organizations. For example, Nissan Motor Co. recently put together a multiteam system to design and manufacture its newest all electric zero-emission vehicle Nissan Leaf (Nissan Design Newsletter). As an illustration of interdependence shared among component teams, the ability of the Leaf multiteam system to meet a new market demand for a longer-lasting battery depended on the engineering team's ability to produce a larger lithium battery as well as on the design team's ability to alter

the design of the car's trunk to accommodate the bigger battery (Murray, 2010; Nissan, 2011). This type of functional interdependence, in which component teams share vested interests and influence each other's work, is a critical feature of multiteam systems (Mathieu et al., 2001).

Research on stand-alone teams suggests that effective planning is an important determinant of team performance (Mathieu & Rapp, 2009; Mathieu & Schulze, 2006; Smith, Locke, & Barry, 1990; Stout, Cannon-Bowers, Salas, & Milanovich, 1999; Weingart, 1992; Weldon, Jehn, & Pradhan, 1991), but planning is likely to be even more critical in a multiteam system context in which collective goals and tactics serve as critical coordination mechanisms (Mathieu et al., 2001). Because multiteam systems are too large to support mutual adjustment among all team members (Davison & Hollenbeck, 2012), planning and goal setting needs to take place within the smaller component teams in which mutual adjustment is possible. In fact, Davison et al. (2012) found that direct mutual adjustment between lower-level members of component teams was actually detrimental to performance in multiteam systems and only served to convert the multiteam system into an overly large, undifferentiated, and unwieldy team. However, simply recognizing the distributed nature of planning in multiteam systems and the limitations of mutual adjustment across all team members does not answer all the procedural questions associated with the planning process. In particular, a question arises as to where the original nexus of planning should be—in the leadership team or the component teams. Multiteam systems could be either centralized or decentralized (Mathieu et al., 2001), yet theory on multiteam systems does not make formal prescriptions about whether centralized or decentralized planning is most appropriate in a normative sense.

Planning is centralized when the formal authority for decision making rests with a leadership team, and planning is decentralized when decision-making authority rests with lower levels of a hierarchy, such as component teams (Mintzberg, 1983). Organizational design literature suggests that planning helps units define a systematic transition from their present state to a desired future state (Galbraith, Downey, & Kates, 2002; Kanter, 1993). Direction of behavior toward the future is attainable through goal setting, which is one of the first steps of planning (Grant & King, 1982; Lorange & Vancil, 1977; Steiner, 1979). In fact, it has been

argued that the planning process is closely intertwined with goal setting in various organizational structures (Galbraith, 1995, 2008; Kanter, 1993; Kates & Galbraith, 2007; Mills & Ungson, 2003). Since design is concerned with creating structures that facilitate goal attainment (Simon, 1981), a theoretically integrative approach to design and goal setting is appropriate when investigating decentralized planning in multiteam systems.

Whereas the literature on small stand-alone teams views decentralizing practices as predominantly positive, theoretical work on large-scale organizations suggests that these practices have both positive and negative features. On one hand, decentralization is considered a solution for problems associated with cognitive limitations, allowing for more planned approaches to decision making, as well as increased motivation levels (Miller, 1987; Mintzberg, 1983; Richardson, Vandenberg, Blum, & Roman, 2002). More specifically, theories of decentralization (Dutton & Duncan, 1987; Miller, 1987; Mintzberg, 1979, 1983; Singh, 1986) suggest that decentralized planning is expected to increase aspiration levels (Kanter, 1993) and proactivity (Bindl & Parker, 2010; Dutton & Duncan, 1987; Parker, Williams, & Turner, 2006).

On the other hand, decentralization may create problems for risk management and knowledge sharing (Mintzberg, 1983). More specifically, researchers have cautioned that decentralizing prac-

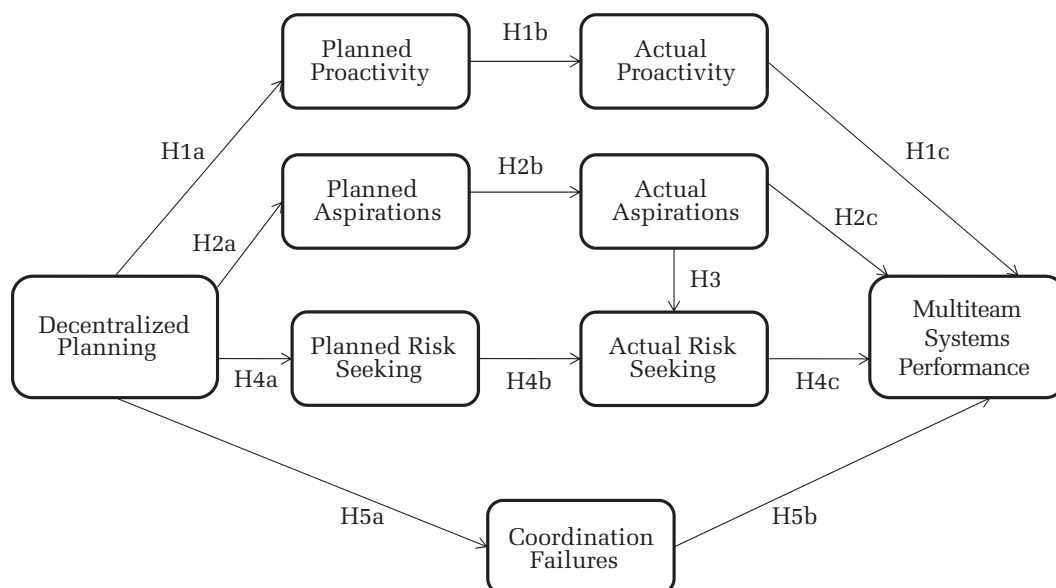
tices may induce excessive risk seeking and coordination problems (e.g., Kanter, 1993; Miller, 1987; Mills & Ungson, 2003; Mintzberg, 1983; Singh, 1986), but these negative effects have not received much attention in the multiteam system literature. This current study integrates theories of decentralization and goal setting and examines the effects of decentralized planning on proactivity, aspirations, risk seeking, and coordination in multiteam systems. Figure 1 portrays our hypothesized model.

HYPOTHESES

Positive Effects of Decentralized Planning via Proactivity and Aspirations

Proactivity. Proactivity has been defined as anticipatory, self-directed, and future-focused action aimed to bring about change (Bindl & Parker, 2010; Grant & Ashford, 2008). It is a way of behaving that involves thinking ahead and taking charge of a situation (Bindl & Parker, 2010) “through scanning for opportunities, showing initiative, taking action” (Kirkman & Rosen, 1997: 152). Structural barriers associated with centralized planning often impede discretion, latitude, and accountability, whereas decentralizing practices create more empowered and autonomous work units that promote flexibility, accountability, and proactivity in fluid and dynamic environments (Kanter, 1993). Research sug-

FIGURE 1
Hypothesized Model



gests that job-enriching interventions that give people more autonomy for planning their own work increase their self-efficacy for engaging in more proactive sets of tasks (Parker, 1998).

Multiteam systems that engage in decentralized planning are likely to set higher goals for proactive behavior and subsequently be more proactive, because they have more control over future events (Zimmerman, 1990), and because they are likely to have higher collective efficacy (Jung & Sosik, 2002). Research has offered some support for these arguments. Williams, Parker, and Turner (2010) studied several self-managed teams that set their own long-term goals and found that self-management was positively related to team proactive performance. Similarly, Kirkman and Rosen (1999) found that organizational policies that gave teams more responsibilities increased their empowerment and subsequently their proactivity. These findings suggest that, relative to more centralized plans, decentralized planning should encourage component teams to have a positive orientation toward their tasks and to be motivated to engage in a broader set of task-relevant behaviors (Kirkman & Rosen, 1997: 1999).

Because component teams are the locus of decision making in decentralized structures, they assume collective responsibility for both goals and multiteam system performance. This collective responsibility is likely to motivate multiteam systems to be more proactive (Fuller, Marler, & Hester, 2006). In keeping with these arguments, we anticipate that when multiteam systems engage in decentralized planning, they will plan to be more proactive and subsequently will behave more proactively by showing more initiative, and by scanning the environment for opportunities that will benefit their performance. Furthermore, proactivity is expected to enhance team performance, because proactive behaviors enable teams to expand their knowledge and to develop skills that contribute to a better understanding of their task environment, allowing them to develop more effective task strategies (Bindl & Parker, 2010; Grant, Gino, & Hofmann, 2011; Grant & Parker, 2009). We summarize our predictions in the following hypotheses:

Hypothesis 1a. Decentralized planning increases planned proactivity.

Hypothesis 1b. Planned proactivity is positively associated with actual proactivity.

Hypothesis 1c. Actual proactivity is positively associated with multiteam system performance.

Hypothesis 1d. Planned productivity mediates the positive effect of decentralized planning on performance.

Hypothesis 1e. Actual proactivity mediates the positive effect of decentralized planning on performance.

Aspiration levels. Theoretical work on decentralized structures suggests that decentralized planning is likely to encourage teams to develop high aspiration levels and set more difficult goals (Kanter, 1993). In contrast, centralized goals set by a leadership team within a multiteam system are likely to reflect lower aspiration levels for two specific reasons. First, Hinsz (1995) has argued that goal setting becomes more and more conservative as groups become larger, due to the effects of anticipated social comparison processes caused by the uncertainty associated with the potential contributions of individual group members. Thus, to provide a “buffer” that would allow for successful goal accomplishment, given the uncertainty caused by their interdependence, groups tend to “discount” their goals. Hinsz (1995) suggested that the larger the group, the larger the uncertainty, and hence the larger the amount of discounting. Because the entire multiteam system is more salient to its leadership team than to its component teams, this discounting effect is likely to be larger for the leadership team, whose members are likely to show lower aspirations if they set and anchor planning and goal setting.

Second, research on goals suggests that commitment to difficult self-set goals tends to be higher than that associated with externally set goals (Latham, Mitchell, & Dossett, 1978). A series of studies by Erez and colleagues (Erez, 1986; Erez & Arad, 1986; Erez & Early, 1987; Erez, Earley, & Hulin, 1985) suggested that people respond better in terms of commitment and implementation to self-set goals than to those set by hierarchical supervisors, and that externally assigned goals have to be “aggressively sold” for them to achieve the same level of acceptance as self-set goals (Latham, Erez, & Locke, 1988). Owing to the lower probability associated with higher goal levels, groups or individuals may perceive difficult goals assigned by others as unfair, and then react by failing to show acceptance or commitment to those goals (Hollenbeck & Klein, 1987).

Thus, there are theoretical and empirical reasons to believe that decentralized planning is likely to impact aspiration levels. In addition, theory and research have suggested that aspiration levels are a primary driver of performance through directing attention, increasing effort and persistence, and leading to use of task-relevant knowledge and strategies (Chesney & Locke, 1991; Latham & Locke, 1991; Locke & Latham, 2002; O'Leary-Kelly, Martocchio, & Frink, 1994; Weingart, 1992). As a result, we posit:

Hypothesis 2a. Decentralized planning increases planned aspirations.

Hypothesis 2b. Planned aspirations relate positively to actual aspirational behavior.

Hypothesis 2c. Actual aspirational behavior is positively associated with multiteam system performance.

Hypothesis 2d. Planned aspirations mediate the positive effect of decentralized planning on performance.

Hypothesis 2e. Actual aspirations mediate the positive effect of decentralized planning on performance.

Negative Effects of Decentralized Planning via Risk Seeking and Coordination Failures

Excessive risk seeking. Multiteam systems exhibit risk-seeking behavior whenever they pursue desired outcomes that are associated with relatively high probabilities of suffering harm or loss. Decentralized planning is likely to have both direct and indirect effects on risk seeking in multiteam systems. First, with respect to indirect effects, research on goal setting theory suggests that when individuals and groups pursue difficult goals, they are more likely to engage in risk-seeking behavior (Heath, Larrick, & Wu, 1999; Larrick, Heath, & Wu, 2009). Although no prior research has studied the effect of difficult goals on risk-seeking behavior in a multiteam system context, studies on stand-alone teams have suggested that this is a possibility. Knight, Durham, and Locke (2001) were the first to study the relationship between goal levels and risk in small teams. They argued that, given an array of strategies with differing risks, teams with high performance goals are most likely to take greater risk, because riskier tactics facilitate the realization of desired performance outcomes. Diffi-

cult goals consistently motivated teams to choose riskier strategies.

Indeed, although difficult goals are traditionally associated with positive outcomes, recent research on the "dark side of goals" has suggested that under certain conditions, high aspiration levels may result in negative side effects. For example, Ordóñez, Schweitzer, Galinsky, and Bazerman (2009) argued that goal difficulty is associated with excessive risk seeking. They posited that difficult goals narrow individuals' focus and cause them to overlook other issues that are important to success. Larrick et al. (2009) found empirically that challenging goals led to riskier strategies in negotiation contexts. They argued that challenging goals motivate strategies that have higher variance but lower expected values than more cautious strategies. Mishina, Dykes, Block, and Pollock (2010) also documented the link between goal difficulty and risk-seeking behavior but took this general idea one step forward by showing how high goals even promoted illegal activity in several organizations. Thus, goals that are too ambitious may create larger negative discrepancies between multiteam systems' actual performance and their aspiration levels. Prospect theory (Kahneman & Tversky, 1979) suggests that these negative discrepancies will further exacerbate risk-seeking behavior. As a result, multiteam systems that seek higher aspiration levels are likely to also exhibit excessive risk-seeking behavior. In keeping with these arguments, we hypothesize:

Hypothesis 3. Pursuing higher aspirations has a positive influence on risk-seeking behavior.

In addition to indirectly affecting risk through aspirations, decentralized planning might directly affect multiteam systems' risk-seeking behavior. Organizational design research has suggested that decentralizing practices that provide employees with more opportunity and authority are also risk-inducing (Hoskisson, Hitt, & Hill, 1991; Kanter, 1993; March & Weil, 2005; Miller, 1987; Singh, 1986). This research suggests that decentralized structures have relatively permissive control systems that favor exploration and risk but overlook costs (Hoskisson et al., 1991; March & Weil, 2005). Decentralized planning, therefore, is likely to encourage higher goals for risk and consequentially excessive risk seeking.

More importantly, this general tendency of decentralizing practices to promote risk seeking is likely to be exacerbated by interteam dynamics in a multiteam system for two reasons. First, group po-

larization effects can take this individual predilection toward risk to a much higher level when people are part of a large group (Cheng & Chiou, 2008; Kameda & Davis, 1990; Myers & Kaplan, 1976). Research has shown that group contexts tend to polarize or shift team attitudes in the direction toward which members are already leaning; hence, the risk-seeking attitude engendered by difficult goals is likely to shift a large group or multiteam system even further in that direction (Buehler, Messervey, & Griffin, 2005; Klein, Dansereau, & Hall, 1994). In addition, the potential for interteam competition between the various component teams in a multiteam system that might not share the same skills and cultures can also produce interteam dynamics that result in enhanced risk seeking due to perceived rivalry (Kilduff, Elfenbein, & Staw, 2010; Malhotra, Ku & Murnighan, 2008).

Second, the tightly coupled nature of multiteam systems has implications for the combined level of risk seeking, which takes on nonadditive properties in multiteam systems. For example, one common decision-making error, referred to as the conjunctive fallacy, is based on the mistaken assumption by naïve decision makers that the combined level of probability for two events most closely resembles the average of those probabilities rather than their product (Tversky & Kahneman, 1983). Costello (2009) has shown how this error results in grossly overoptimistic prediction and planning at the individual level, and the potential for this problem to be exacerbated at the multiteam system level cannot be discounted. Thus, if one component team is pursuing a risky strategy that has only a 20 percent chance of success, and another component team is also pursuing a risky strategy with only a 20 percent chance of success, the overall combined probability of success is less than 5 percent. For all these reasons, we expect multiteam systems that engage in decentralized planning to set higher goals for risk seeking and exhibit excessive risk-seeking behavior that will hurt multiteam system performance (Bromiley, 1991; Wiseman & Bromiley, 1996). Thus, we posit:

Hypothesis 4a. Decentralized planning increases planned risk seeking.

Hypothesis 4b. Planned risk seeking is positively associated with actual risk seeking.

Hypothesis 4c. Actual risk seeking is negatively associated with multiteam system performance.

Hypothesis 4d. Planned risk seeking mediates the negative effect of decentralized planning on performance.

Hypothesis 4e. Actual risk seeking mediates the negative effect of decentralized planning on performance.

Coordination failures. Coordination between teams is critical to multiteam systems' success, because component teams are functionally interdependent, depending on each other for accomplishing a common objective too large to be achieved by a single team (Hoegl et al., 2004; Marks, DeChurch, Mathieu, Panzer, & Alonso, 2005). In centralized planning, their leadership team decides on multiteam systems' strategies, their timing of activities, and the synchronicity of component team actions (DeChurch & Marks, 2006). In decentralized planning, however, leaders take a more passive role by coordinating decisions made by component teams. Because of the different component teams involved in decision making and their specialized functions, decentralized planning is likely to introduce coordination complexities for a multiteam system. Whereas ineffective coordination may not harm the effectiveness of within-team dynamics in the component teams, it could harm the success of the larger multiteam system because the large size of the multiteam system and skill-based differences between component teams may preclude effective, direct interteam communication and coordination (e.g., DeChurch & Marks, 2006).

The potential for decentralization to create system-level coordination problems (redundancies and cannibalization) has been documented in the organizational theory literature on large-scale organizations in which centralization is often considered a source of efficiency (Drazin & Van de Ven, 1985; Leavitt, 2005; Magee & Galinsky, 2008; March & Simon, 1958; Thompson, 1967). These findings from studies of large-scale organizations may not be relevant to traditional teams, because the small size of these teams allows for direct, face-to-face mutual coordination in real time; however, they are very germane to larger multiteam systems, in which direct mutual coordination in real time between all team members actually harms performance (e.g., Davison et al., 2012). Thus, plans and goals set independently by decentralized component teams may result in plans and goals that are less well coordinated than what might result from centralized plans and goals set by a leadership team, which could create coordination failures in which

the actions of the component teams are neutralized or counteracted by the leadership team or other component teams. Given these arguments, we propose:

Hypothesis 5a. Decentralized planning increases coordination failures.

Hypothesis 5b. Coordination failures negatively affect performance.

Hypothesis 5c. Coordination failures mediate the negative effect of decentralized planning on performance.

Although it is possible to leverage existing theory and data to generate specific positive and negative paths by which decentralized planning might affect multiteam system performance, the lack of direct data on multiteam systems makes it difficult to speculate on whether the two different sets of effects are likely to neutralize each other or whether one set of effects will dominate. For this reason, we make no direct prediction about the overall impact of these effects and instead state this research question: Will the combined set of positive and negative effects associated with decentralized planning result in an overall positive effect, neutral effect, or negative effect in multiteam systems?

METHODS

Our study sample consisted of 2,940 United States Air Force officers from a broad array of job categories, who participated in a five-week leadership development course. As part of the course and a larger program of research,¹ the participants were assigned to one of 210 14-person multiteam systems that engaged in a computer-based simulation. Each multiteam system engaged in the same task, received the same training, and had the same time limitations. The teams were comparably composed in terms of age, experience, gender, occupational category, and Armed Services Vocational Aptitude Battery (ASVAB) scores to ensure that they had equal opportunity to perform well on the simulation. The participants were motivated to take the task seriously, because performance on the simulation was part of an overall evaluation that affected

their future advancement and opportunities for promotion.

Our research design represents a situated experiment that optimizes both the traditional strengths associated with laboratory research and field research (Greenberg & Tomlinson, 2004). The setting was lablike in terms of standardization, control, and manipulation of conditions. At the same time, the context was fieldlike because research participants were experienced decision makers working at a task that was similar to what they did or supported in their real jobs in a context in which performance had important career consequences.

Task

We used the Leadership Development Simulation (LDS), a computer-based simulation, as an experimental platform for this study. The LDS was jointly developed by the United States Air Force and a large research-oriented university and has been used in other studies on multiteam systems (see Davison et al., 2012). The simulation, although laboratory-like, had face validity for this sample, because it involved the management of remotely piloted aircraft (RPAs), which are assets used to search and engage enemy targets in a dynamic environment and reflect an innovation that was being incorporated by this organization. The simulation was a relatively unstructured task that presented an initially unknown dynamic environment to teams that had to discover and engage hidden targets. The game grid consisted of a 16 by 16 matrix and contained different types of hidden targets (see Figure 2). The goal of the multiteam system in this simulation was to employ operational and intelligence assets to find and engage all types of hidden targets. More information about the targets is contained in the Appendix.

Multiteam System Structure and Team Roles

As depicted in Figure 3, each multiteam system was composed of three component teams labeled the Point Team (operations), Support Team (intelligence), and the Leadership Team. All teams were composed of six members, but because the directors and assistant directors were boundary spanners who were part of both their multiteam system's leadership team and one of its component team, the overall size of each multiteam system was 14 persons. The Point Team was responsible for finding and engaging targets in the game board by

¹ Fifty-nine percent of the teams that made up the Davison et al. (2012) sample also participated in this sample. Whereas there is no overlap in independent or mediating variables across these two studies, future meta-analyses should recognize the nonindependent nature of the performance measure in these samples.

FIGURE 2
Screenshot of the LDS Grid

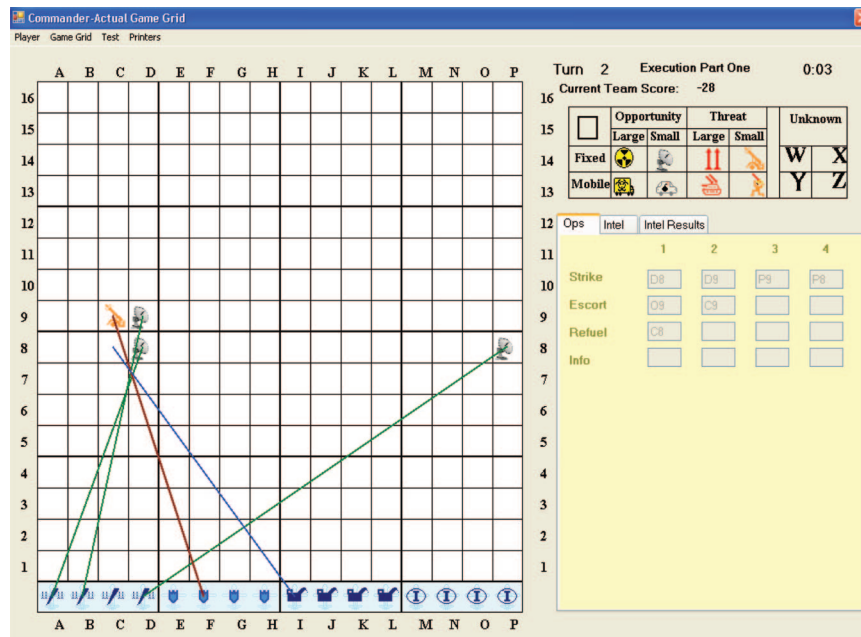
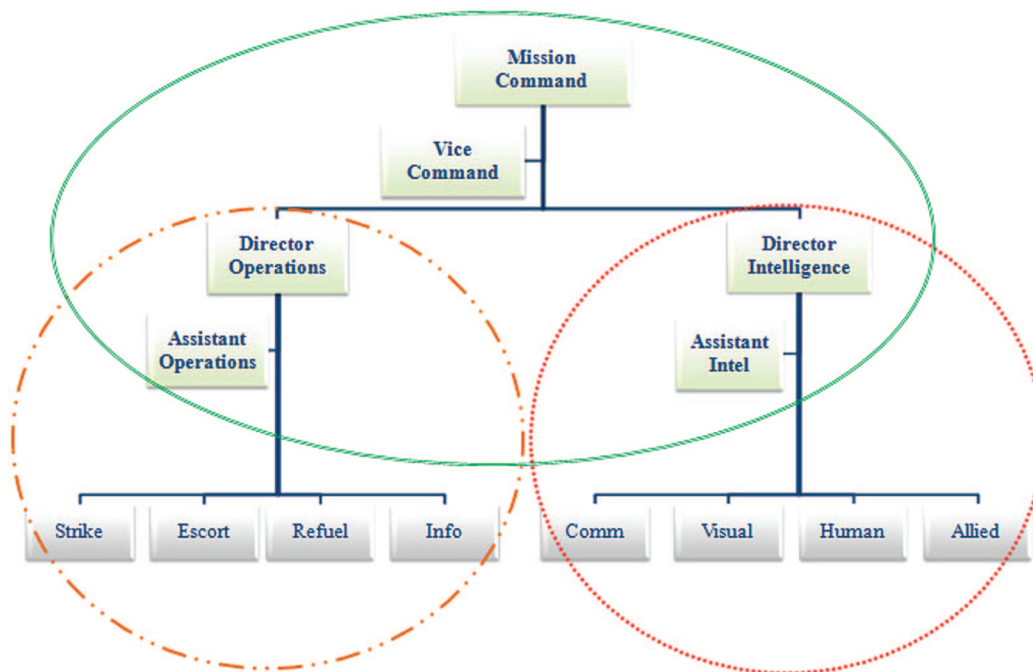


FIGURE 3
Structure of Multiteam Systems^a



^a The circle outlined solid (—) represents the six-member Leadership Team, including the directors and assistant directors, who were boundary spanners between the Leadership Team and each of the component teams. The circle outlined with dots and dashes (— · · —) represents the Point (Operations) (Component) Team, and the circle outlined with dots only (·····) represents the Support (Intelligence) (Component) Team.

using four different types of RPAs that had specific and unique capacities. The Support Team was in charge of four different types of intelligence assets, and its main function was to provide information about the location of hidden targets. The Leadership Team provided overall guidance for the multiteam system. The Appendix contains more information about the component teams and their roles.

Role enactment. The simulation scenario lasted approximately three hours, not counting training. Lower-level staff members, directors, and the mission commander had “rolling access” to the simulation, which allowed us to time-stamp and capture specific decisions made at specific times. The teams made anywhere between 1,500 and 2,000 decisions over the course of the whole simulation, depending upon how they approached this unstructured task. All of these decisions were captured, coded, and time-stamped by the simulation for members of the 210 14-person multiteam systems that participated in this research. All 14 team players could see the results of operational missions on a large master screen in the room, and the results of intelligence assets in printouts produced by the simulation.

Training

Training on the functionalities of the simulation consisted of two parts. In part one, team members watched a 45-minute video training that explained the basic functionalities of the simulation, such as its purpose, the nature of the targets, roles and responsibilities, and other simulation-pertinent functionalities. In part two, multiteam systems participated in a 15-minute hands-on training session, during which team members interacted with a shorter version of the actual simulation. The hands-on training session simulated different scenarios that multiteam systems could face during the real simulation (such as sending the wrong asset to the wrong targets). Team members were instructed to move targets on the board and encouraged to ask questions. The hands-on training session was followed by an extensive practice session for all teams. Multiteam systems played the simulation in a functional structure (Hollenbeck et al., 2002) in which each player was responsible for one type of asset.

Manipulation and Measures

Decentralized planning. As we noted earlier, the large size and specialized nature of component

teams makes it neither feasible nor desirable for all members of the multiteam system to meet face-to-face in real time to plan and make decisions. Instead, planning and goal setting in multiteam systems needs to be distributed and thus take place within the smaller teams (i.e., the component teams or the leadership team) where mutual adjustment is feasible. However, because the directors and assistant directors cannot be in two places at once, the result is a sequential planning process in which some teams go first and others go second.

We manipulated decentralized planning through a presimulation planning and goal-setting session that gave initial planning authority to component teams in the decentralized planning condition or to the leadership team in the centralized planning condition. In the decentralized condition, the directors and assistant directors worked first with their lower-level staffs (operations or intelligence) to come up with an initial plan that was then presented to the leadership team by the boundary-spanning directors and assistant directors. In contrast, in the centralized condition, the directors and assistant directors worked first with the leadership team to come up with an initial plan that was then presented to the component teams by the boundary-spanning directors and assistant directors. As part of the planning process, the multiteam systems set goals for proactivity, aspiration levels, and risk seeking.

Planned and actual proactivity. Multiteam systems sent their intelligence assets out in attempts to discover targets hidden in any of the 256 squares of the game grid, which could be probed by sending either a single-sourced mission (e.g., only one type of query was made within that space) or a double-sourced mission (e.g., two different queries were made within the same space). During planning, multiteam systems set goals for the percentage of single-sourced missions that they would send out on the game board during the simulation. The intent to send out more single-sourced missions reflected multiteam systems' desire to emphasize the scope of activity in order to find and destroy more targets. Hence, we operationalized *planned proactivity* as the percentage of single-sourced missions that multiteam systems planned to send out on the game board. To facilitate the decision-making process for the teams, we converted the percentage of planned single-sourced missions into a Likert scale (1 = 0%, 5 = 100%). Teams discussed the five different strategies and then selected one; this measure was used to operationalize planned proactiv-

ity. When multiteam systems sent out a large number of intelligence missions, the measure showed that they proactively covered more space on the game board to find targets and score higher points. Hence, we operationalized *actual proactivity* as the total number of intelligence missions sent out throughout the game. Both planned and actual proactivity represent multiteam systems' desires to contribute to multiteam system performance by actively exploring and engaging the game grid to score more points. These intentions and subsequent behaviors capture the anticipatory and change-oriented nature of proactivity (Grant & Ashford, 2008).

Planned and actual aspirations. Each multiteam system was provided with an inventory of possible targets and a summary of the total number of points available if all of the targets were destroyed. They were also given normative information about how various raw scores from targets destroyed translated into percentiles of performance. The multiteam system then set a goal for the number of points they were going to try and accomplish. We operationalized *planned aspiration levels* as the degree to which the goals set were high versus low in terms of normative probabilities. The normative probabilities varied along a range in which 1 corresponded to the top 10th percentile and 5 to the bottom 10th percentile. Multiteam systems referred to these normative scores to decide on an overall aspiration score; this measure was used to operationalize planned aspiration levels.

Participants were also told within the pregame debriefing that most of the high-value opportunities were within well-protected spaces in the northern region of the task environment but that they could expect base attacks launched in the southern region. Because more points were associated with prosecuting high-value targets than with having bases hit, teams that had high aspirations had to send more missions to the northern region of the task space to score more points. In addition, prosecuting well-protected high-value opportunities in the northern region required a large, coordinated package of RPA missions that had to include "strikes," "escorts," and "refuelers," and thus limited the number of missions that could take place in other regions. Thus, we operationalized *actual aspirational behavior* as the number of missions that were actually flown to the northern region of the task space in pursuit of high-value targets.

Planned and actual risk seeking. Escort RPAs could protect other RPAs from being destroyed. Thus, any mission that was double-escorted was totally safe and risk-free, and any mission that was unescorted or single-escorted was at risk. The multiteam systems set goals for how risk seeking they were going to be in terms of what percentage of missions was going to be double-escorted. If a multiteam system planned on conducting 100 percent, double-escorted missions, this was very risk averse. In contrast, a multiteam system's planning on conducting 0 percent double-escorted missions was very risk seeking. To facilitate the decision-making process for the teams, we converted the different possible double-escorted strategies into a Likert scale where 1 corresponded to 100 percent of double-escorted missions (resulting in 20 overall double-escorted missions) and 6 corresponded to 0 percent of double-escorted missions (resulting in 160 total missions not double-escorted). Teams discussed the different strategies and then selected one; this measure was used to operationalize planned risk seeking.

In terms of actual behavior, any mission that was not escorted entailed some risk, but this risk was small if the team knew what was in the space from prior missions or validated intelligence reports. In contrast, risk seeking was only high when a multiteam system sent an unescorted mission into blind space, that is, a space into which the team had never gone before with either an operational or intelligence asset. Thus, the actual number of unescorted missions sent into blind space was then used as the operationalization of *actual risk seeking*.

Coordination failures. Obviously, no multiteam system plans on being poorly coordinated, so we did not have multiteam systems set goals for how poorly they were going to be coordinated. We did measure the actual level of *coordination failures*, however, in terms of (a) the frequency of editing behavior on the part of the leadership team that resulted in negative outcomes for the multiteam system (vertical coordination) and (b) the frequency of missions conducted by the operations team that were not based upon the multiteam system's shared understanding of the task environment or common operational picture (COP) (horizontal coordination). Vertical coordination failures occurred whenever a leadership team (a) moved an RPA that was going to a space where it would have been safe to a space where it was instead destroyed or (b) edited a revealing intelligence query by de-

letting it or moving it to an empty space. In each case, these actions were an indication that the Leadership team's understanding of the situation was not well aligned with that of the component teams in ways that degraded the overall performance of the multiteam system.

Horizontal coordination failures occurred whenever component teams conducted operations that were not based upon information that resided on the COP, an up-to-date mental picture of the battle space projected on a large screen for all team members to see. A large number of COP-based missions meant that the point team and the support team were well coordinated, but a large number of non-COP-based missions reflected the fact that the point team and the support team were not working together. Teams that were poorly horizontally coordinated were characterized by (a) point teams that based their decisions on their own experience and ignored the COP and (b) support teams that were only concerned with placing valid information on the COP but ignored whether that information was ever used for actual mission targeting. These coordination failures indicated that the component teams were more preoccupied with local issues associated with their own component team performance than with the performance of the multiteam system as a whole. The simulation recorded these failures, and we standardized variables because of differing measurement units and then took their average as the overall index of coordination failures.

Multiteam system performance. This was an overall objective measure equal to the sum of points that multiteam systems gained by destroying

targets minus points lost from destroyed assets and hits registered on their bases.

RESULTS

The descriptive statistics, including means, standard deviations, correlations, and sample size are presented in Table 1. We had missing data for some teams because goals were derived from the planning worksheets, and these were not always filled out completely. We tested the degree to which missing data were random or systematic by examining means and standard deviations for measures of teams with complete data with the means and standard deviations of teams that had missing data. No statistical differences were found between samples.

To test the proposed model, we used structural equation modeling using maximum likelihood estimation in MPLUS 4.2 (Muthén & Muthén, 2006). MPLUS uses full information maximum likelihood as a default that deals with missing data while using all available data points to estimate model parameters. We used the following criteria to assess model fit: chi-square with probability greater than .05; comparative fit (CFI) and Tucker-Lewis index (TLI) values greater than .90, a root-mean-square error of approximation (RMSEA) of less than .06, and a standardized root-mean-square residual (SRMR) of less than .08 (Hu & Bentler, 1999). The overall model had good fit to the data ($\chi^2[24, n = 210] = 33, p = .11$; CFI = .94, TLI = .92, RMSEA = .04, SRMR = .055). Path coefficients are presented in Figure 4.

TABLE 1
Descriptive Statistics and Correlations^a

Variable	Mean	s.d.	1	2	3	4	5	6	7	8
1. Decentralized planning	0.60	0.50								
2. Planned proactivity	3.10	1.09	.13							
3. Actual proactivity	166	69.4	.20*	.30**						
4. Planned aspiration levels	180	28.0	.24**	.05	.13					
5. Actual aspiration levels	25.5	5.48	.06	.04	.07	.15*				
6. Planned risk seeking	2.80	1.18	.32**	.20*	.05	.12	.16*			
7. Actual risk seeking	28.8	13.8	.13	.06	.01	.02	.30**	.32**		
8. Coordination failures	0.00	2.30	.23*	.08	-.05	.12	-.15*	.02	.02	
9. Multiteam system performance	151	43.4	-.14*	-.11	.14*	-.01	.16*	-.05	-.21**	-.43**

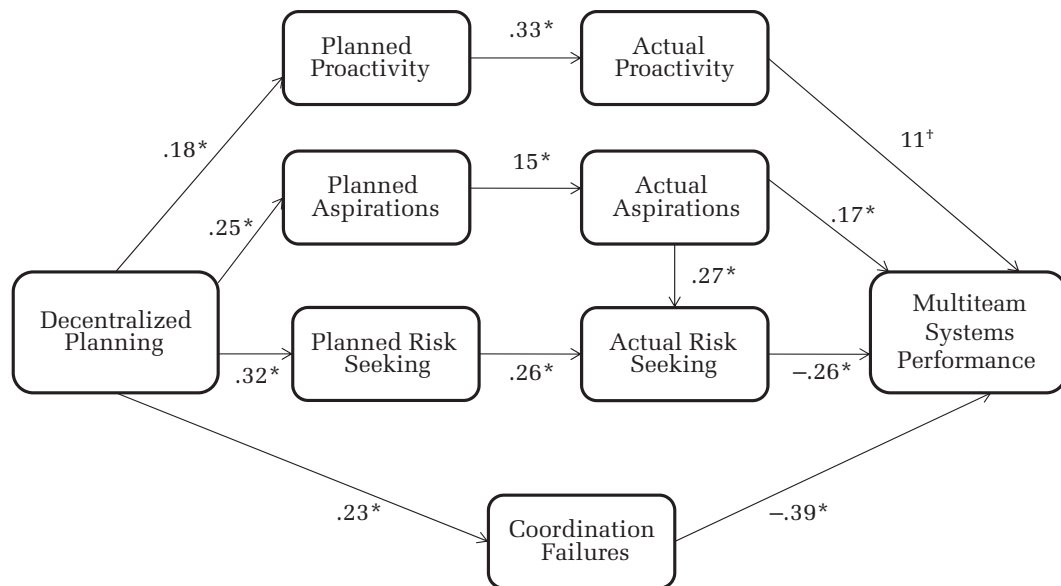
^a Standardized regression coefficients are reported. The sample size for the multiteam systems with no missing data on any one of the planned goals varied from 167 to 210.

* $p < .05$

** $p < .01$

Two-tailed tests.

FIGURE 4
Tested Model^a



^a Results of tests of the hypothesized model; $n = 210$.

* $p < .05$

† $p < .10$

Since we propose a three-stage mediated model, we used procedures developed by Taylor, MacKinnon, and Tein (2008) to examine our mediation hypotheses. We performed a bias-corrected bootstrap analysis to calculate the confidence intervals (CIs) for the mediated (indirect) effects. We found partial support for Hypothesis 1. Specifically, we found that multiteam systems that engaged in decentralized planning planned for higher proactivity (Hypothesis 1a) ($\beta = .18$, $p < .05$), which in turn was associated with higher actual proactive behavior in the simulation (Hypothesis 1b) ($\beta = .33$, $p < .05$). However, performance of proactive behavior was only marginally related to performance (Hypothesis 1c) ($\beta = .11$, $p < .10$). The indirect effect of decentralized planning on performance through goals for proactivity and subsequently proactive behavior was significantly different from zero (95% CI: lower bound = 0.01; upper bound = 1.69), thus providing evidence for mediation (Taylor et al., 2008) and support for Hypotheses 1d and 1e.

Supporting Hypothesis 2, decentralized planning predicted higher planned aspiration levels (Hypothesis 2a) ($\beta = .25$, $p < .05$), which in turn was associated with more aspirational behavior (Hypothesis 2b) ($\beta = .15$, $p < .05$), which then was positively related to multiteam system performance (Hypothesis 2c) ($\beta = .17$, $p < .05$). The mediational

expectations (Hypotheses 2d and 2e) were also supported, as the indirect effect of decentralized planning on multiteam system performance via aspirations was significant (95% CI: lower bound = 0.05; upper bound = 1.8). Thus, in line with our integrated theoretical perspective (Kanter, 1993; Latham & Locke, 1991; Locke & Latham, 2002; Mintzberg, 1983), we found that decentralized planning had positive effects on multiteam system processes (higher levels of planned and actual proactivity and aspirations), which in turn had positive effects on multiteam system performance.

In contrast to the positive effects of decentralized planning on multiteam system performance via proactivity and aspirations, however, we found that decentralized planning had negative effects on multiteam system performance attributable to risk seeking. Thus, pursuit of higher aspirations was positively associated with more risk-seeking behavior ($\beta = .27$; $p < .05$), in support of Hypothesis 3. Consistently with Hypothesis 4, decentralized planning also resulted in higher planned risk seeking (Hypothesis 4a) ($\beta = .32$; $p < .05$), which was then associated with higher levels of actual risk-seeking behavior (Hypothesis 4b) ($\beta = .26$; $p < .05$), which in turn negatively related to multiteam system performance (Hypothesis 4c) ($\beta = -.26$; $p < .05$). Mediational analyses were supported—

planned and actual risk-seeking behavior mediated the effect of decentralized planning on multiteam system performance (Hypotheses 4d and 4e; 95% CI: lower bound = -4.2 ; upper bound = $-.7$).

In keeping with Hypothesis 5, we also found that decentralized planning had negative effects on performance attributable to coordination failures. Decentralized planning resulted in greater frequency of coordination failures among teams (Hypothesis 5a) ($\beta = .23$; $p < .05$), which in turn was negatively associated with multiteam system performance (Hypothesis 5b) ($\beta = -.39$; $p < .05$). Mediation was also present, as the indirect effect of decentralized planning on performance via coordination failures was significantly different from zero (Hypothesis 5c: 95% CI: lower bound = -14 ; upper bound = -3). This finding is particularly relevant in light of recent work suggesting that horizontal and vertical coordination contributes positively to multiteam system performance (Davison et al., 2012).

Finally, with regard to our research question, we found that the overall effect of decentralized planning on multiteam system performance was negative. The zero-order correlation between decentralized planning and overall multiteam system performance was statistically significant ($r = -.14$, $p < .05$). This effect was mediated by the positive and negative effects of decentralized planning on multiteam system performance, and the direct effect of decentralized planning on multiteam system performance was reduced to nonsignificance ($\beta = -.06$; n.s.) when the hypothesized paths were included in the path model. The total effect of decentralized planning on multiteam system performance through planned and actual behaviors was $-.10$ ($p < .05$), and the hypothesized model explained 24 percent of the variance in performance.

DISCUSSION

This study challenges the degree to which past theories and findings regarding traditional teams, with their emphasis on within-team dynamics, generalize to multiteam systems that instead place a priority on between-team dynamics. We investigate this possibility for decentralized planning in multiteam systems by utilizing an integrative framework of theories of decentralization, goal setting, and proactivity.

Theoretical Contributions

Traditional team-based theories suggest that decentralized structures have predominantly positive

effects for stand-alone teams. Our present work challenges these theories. In keeping with research on decentralizing interventions in small stand-alone and empowered teams (e.g., Chen et al., 2011; Cordery et al., 2010; DeJoy et al., 2010), we found that decentralized planning had positive effects on multiteam system performance due to higher proactivity and higher aspiration levels. However, in contrast to past research that has only examined results at the team level, we found that decentralized planning also had negative effects attributable to between-team dynamics, especially as these are manifested in enhanced risk seeking and interteam coordination failures. Although we made no formal prediction regarding the overall effects of decentralized planning, we found that its negative effects on performance were actually stronger than the positive effects. This could be because performance in multiteam systems is nonadditive in nature. Thus, performance of each component team is limited by the performance of other component teams due to the conjunctive properties of multiteam system activities. Although our findings do not refute the validity of theories that promote decentralization, they do place an important boundary condition on their application to multiteam systems, which are an increasing occurrence in organizations (Haas, 2010; Zaccaro et al., 2012).

Furthermore, our work advances present knowledge and theory about multiteam systems. Recent work on multiteam systems has shown that, in general, coordination through unbridled mutual adjustment among all team members is detrimental to multiteam system performance (Davison et al., 2012). Research has indicated that horizontal and vertical coordination enacted by boundary spanners is critical to multiteam system performance, and that team members need to channel their input through the boundary spanners instead of communicating directly outside the formal structure. The current study did not focus on communication during the task, but rather on planning prior to task engagement, and specifically, on whether the planning process should be decentralized (driven by component teams and their boundary spanners working together first) or centralized (driven by leadership team and boundary spanners working together first). In addition, it examined variables that resulted from the planning manipulation such as aspirations, risk, proactivity, and coordination failures; these variables have never been studied before in a multiteam system context. This work, therefore, extends previous research (e.g., Davison

et al., 2012) by testing whether knowledge about decentralization—a well-researched practice in the small-team literature—ought to be generalized to multiteam systems.

Our research also advances knowledge on proactivity and goal-setting processes in teams. To our best understanding, this is the first study that has taken a process view of both proactivity (e.g., Grant & Ashford, 2008) and goal setting (e.g., Gollwitzer, 1999) in an interdependent team setting. Grant and Ashford (2008) argued that proactivity is a goal-driven process and that planning and action are two important phases in the process. We captured these phases by differentiating between goals for proactivity and actual proactive behavior in multiteam systems. This differentiation allowed us to go beyond simply explaining variance in performance to clarifying how proactivity unfolds as a process in multiteam systems (e.g., Grant & Ashford, 2008). Similar to proactivity, goal setting is also a process in which goals activate task knowledge that in turn enables better performance (Locke, 2000). The goal-setting process has two main phases: goal setting and goal striving (Gollwitzer, 1999). We portrayed this process by capturing both planned activities and actual behavior, which map conceptually into goal setting and goal striving, respectively. We believe that distinguishing between the interrelated planned and actual constructs is particularly relevant when implementing process theories of management and organizations such as goal setting and proactivity (e.g., Suddaby, 2010).

Empirical and Practical Contributions

Empirically, our study contributes to literature on teams and multiteam systems in two ways. First, it is unique because we manipulated decentralized planning and then measured its effects on multiteam systems' dynamics and performance via objective measures of real behavior in a standardized task environment. In addition, we separated measures in time and source, thus minimizing problems of common source and method biases (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003), as well as eliminating the possibility of reverse causality.

Second, this study employed a large number of relatively large multiteam systems. To date, the general tendency in studies examining multiteam systems in laboratory contexts has been to study multiteam systems that are so small (four to six members) that they begin to look and behave like stand-alone teams. DeChurch and Marks (2006) al-

luded to this problem and encouraged more research on larger, more specialized multiteam systems. At the same time, the tendency in field studies of multiteam systems has been to study a very small number of idiosyncratic and noncomparable multiteam systems (e.g., Hoegl et al., 2004). In contrast, we studied a large number of large multiteam systems using a situated experimental research design in which all multiteam systems had the same standardized task, resources, personnel types, and time constraints. Moreover, in this study's context, participants were experienced decision makers working on a task that was representative of the mission of their organization and for which in-role performance had personal career consequences. This is a unique configuration with no real parallel in the multiteam systems literature; hence, the empirical evidence we provide in this study is also unique.

Our study also has several practical implications. With respect to planning in component teams, leaders of component teams need to recognize that the probability of goal accomplishment at the multiteam system level may be a nonadditive function of the probabilities associated with component team goals, and hence they need to guard against overly optimistic planning. With respect to planning in the leadership team, our research shows that centralized planning processes reduced the number of coordination failures between component teams. Thus, multiteam systems may need to place more emphasis on a priori planning, because the large number of people in a system may preclude real-time communication between all members. In addition, language or cultural differences between members of different specialized teams may make effective communication difficult—especially if component teams become preoccupied with their own proximal goals.

This study may also have implications for applied organizational interventions based on theories of empowerment because decentralization is often described as a critical element of empowering interventions (e.g., Aryee et al., 2012; Biron & Bamberger, 2010; Mathieu, Gilson, & Ruddy, 2006; Seibert et al., 2011). The typical recommendation in this literature to decentralize teams (e.g., Hooker & Csikszentmihalyi, 2003; Pearce & Sims, 2002) may not be warranted when these teams are embedded in larger, interdependent multiteam systems. As our results indicate, functional interdependencies and complexities in multiteam systems may offset some of the virtues of decentralized struc-

tures that have been documented in traditional stand-alone teams.

Limitations and Directions for Future Research

While there are liabilities associated with retrospective self-reports and virtues associated with objective measures, no measure is a perfect representation of its underlying construct. This multiteam system context limited the nature of the measurement in some ways; thus, there is a need for additional research using other measures to replicate and extend the findings reported here. For example, although we directly assessed goals and actual behaviors related to proactivity, risk seeking, and coordination failures, we were unable to measure any of the cognitive or emotional variables that could be driving those behaviors. In addition, the wide scope and sheer size of a research effort with this many large multiteam systems precluded direct observation or extensive surveying of teams, because 15–20 of the 14-person multiteam systems were typically run simultaneously.

In addition, because this study was conducted with a simulation that was run in a single organizational context, there are the traditional concerns regarding the generalizability of these findings to other organizational contexts. Although this study meets the evaluation criteria often suggested for laboratory studies (e.g., use of behavioral measures, task psychological realism, and theoretical contribution [Colquitt, 2008]), future research should investigate the effect of decentralized planning in other contexts and with other multiteam systems. Our sample consisted of Air Force officers, who may have been trained to be more responsive to centralized planning processes. The generalizability of the findings reported here, therefore, will become more evident as other researchers, perhaps interested in these findings, replicate this study with other experiments employing different samples and tasks conducted at different times.

Our research suggests several possibilities for future empirical studies on multiteam systems. For example, we employed a relatively strict functional structure (Hollenbeck et al., 2002); however, there is a continuum of organizational designs that fall between highly interdependent functional structures and less interdependent structures (Galbraith, 1971). Thus, it may be the case that the negative effects attributable to decentralized planning here may be reduced or reversed in more divisional structures. For instance, in this context, rather than

creating two highly differentiated and specialized component teams (operations and intelligence), the organizational structure could have created eastern and western divisions, each with its own independent operations and intelligence units. Such a divisional structure might create less interdependence and reduce some of the negative effects of decentralization. Of course, at the extreme, once the level of interdependence between component teams becomes zero, the teams are by definition no longer part of a multiteam system and revert to stand-alone teams with the traditional expectations regarding the virtues of decentralized planning.

Future research is also needed to test if the negative effects associated with decentralized planning might dissipate over time. For example, multiteam systems that utilize decentralized planning may learn from their past mistakes and may, in turn, engage in better coordination and become more risk-averse as they accumulate trial-and-error experience. This issue is somewhat complex, because whereas some evidence supports this possibility (e.g., Haleblan, Kim, & Rajagopalan, 2006), theory and research also suggest that poor initial performance may promote even greater risk seeking. Indeed, a large body of research utilizing prospect theory (Kahneman & Tversky, 1979) and the behavioral theory of the firm (Argote & Greve, 2007; Cyert & March, 1963) has shown that when organizations perform below their aspiration levels, they respond by taking more risk in the future and often perform even worse as a result (e.g., Bromiley, 1991; Fiegenbaum & Thomas, 1988; Kim, Haleblan, & Finkelstein, 2011; Miller & Chen, 2004). It is plausible, therefore, that subpar performance may persist in decentralized multiteam systems. Future research ought to investigate these possibilities.

There are theoretical reasons to suspect that the adverse consequences of decentralized planning may be amplified in settings where multiteam systems' tasks are complex or inherently risky. Complex tasks require more coordination among team members, because each person needs to process more informational cues and resolve more dependency issues (Espinosa, Slaughter, Kraut, & Herbsleb, 2007). A priori planning is necessary for complex tasks; hence, any detrimental effects that are attributable to planning failures caused by poorly integrated decentralized decision making are likely to manifest more clearly for complex tasks. Similarly, decentralized planning could have stronger negative effects in uncertain and riskier contexts

than in contexts in which it is easy to predict outcomes. This potential might be especially relevant in multiteam system contexts, because decisions and behaviors of a large number of people can impact the outcomes associated with one's own decisions and behaviors. Future research should explore if the findings reported here are moderated by task complexity, as well as uncertainty and risk seeking associated with decision-making outcomes.

Team factors such as size, diversity, history and expertise, and physical distance may also moderate some of our findings. For example, decentralized planning may become even less effective for large multiteam systems that are dispersed geographically, since this intensifies coordination and communication complexities. Similarly, functional and cultural diversity may also hinder the understanding of goals and plans and their effective communication across different component teams. Conversely, decentralized planning may work better for teams that have a long history together, and that have developed routines and trust. Exploration of such possibilities would enrich our understanding of multiteam systems and the way they plan.

Clearly, both expanded theory building and empirical research are needed to more fully elucidate the knowledge base regarding multiteam systems. Multiteam systems are neither traditional teams nor standard large-scale organizations; thus, theories and empirical findings from these traditional literatures may not generalize to multiteam system contexts. Multiteam systems are larger and more specialized than traditional teams but smaller and less formalized than large-scale organizations, creating a potential organizational form that can manage a great deal of complexity, yet remain responsive in real time. Realization of this potential, however, is not guaranteed, and we hope that the research reported here will serve as a catalyst for others to explore the dynamics of multiteam systems. Such exploration is clearly needed, because there is a dearth of research on multiteam systems, and as our study suggests, one simply cannot assume that what is true for traditional small stand-alone teams is true for multiteam systems.

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APPENDIX

Target Set in LDS

Operations assets. There were two types of targets hidden in the game grid—threats and opportunities. In addition, targets could be fixed or mobile. Threats were dangerous because sending the wrong assets to threats resulted in destroyed assets and lost points. In addition, mobile threats could move toward the base of the simulation, attack the base, and cost points to the multiteam system. Sending the wrong assets to opportunities, on the other hand, limited the range of missions that the multiteam system could perform, but it did not result in lost points. Opportunities were harmless in terms of losing

points but were the major source for gaining points. The value of targets varied such that destroying large opportunities was worth more performance points than destroying other types of targets. Successful performance on the simulation required multiteam systems to be efficient in utilizing their assets to locate and destroy targets on the game board, to closely monitor their bases, and to avoid base attacks.

Intelligence assets. There were four different intelligence assets: human intelligence, allied intelligence, visual imagery, and communication intercepts. Prior to the simulation, teams in both conditions were told that one intelligence asset would be accurate in the north region (rows 9–16), one would be accurate in the south region (1–8), one would be accurate in the central region (5–12), and one would be accurate in the border regions (1–4 and 13–16). The goal of the intelligence team was to find which asset was accurate in which region, and then cover that region with queries that would support future operations. Each asset had a 95 percent accuracy probability in their “sweet spot” (game-grid area where they were accurate) and a 5 percent accuracy rate in non-sweet spots (other sections in the game grid). If assets were sent to non-sweet spots they would report back inaccurate information 95 percent of the time.

Multiteam systems could verify sweet spots in many different ways. They could use operation assets to test ideas generated by past intelligence missions. For example, if a nonvalidated intelligence source suggested that there was a large fixed threat in a specific cell, the operations unit could send a pair of escorts to that cell to see if that was true. In addition, Operations and Intelligence team players could each send an asset to the exact same place, and the Intelligence Team player could check the accuracy of his or her source against what the operations person saw. In addition, two Intelligence Team players could send different sources to the exact same place in the same round (or adjacent rounds) to see if they could get two confirmations of the same finding. Also, the Operations Team could identify fixed targets and leave them up (as opposed to engaging them right away) so that intelligence staff could test the accuracy of alternative sources with respect to known targets. This requires some degree of patience of the Operations Team, in the sense that members need to forego the potential for immediate points in order to validate intelligence sources, which might pay off in the longer run. There are probably other means for doing this well, but this was a relatively complex task, in the sense that with four sources and four regions, 16 different hypotheses had to be tested by the team, and the more the team worked together, the more likely they were to deduce the best source-region match.

Component Teams and Their Roles

The Point Team (Operations). Each of the four lower-level members of this component team had the ability to

launch one specific type of RPA that included (a) information RPAs, (b) refueling RPAs, (c) strike RPAs, and (d) escort RPAs. Information RPAs were used to discover hidden targets on the game grid; refueling RPAs were used to enable missions sent to the northern half of the game grid; strike RPAs destroyed opportunity targets; and escort RPAs destroyed threat targets. Each member of the operations team controlled four RPAs; hence, the multiteam system as a whole could utilize a total of up to 16 operations assets per game round (four per asset type). The director of operations and assistant director of operations evaluated the missions of their component teams and could assist their component team by making additional moves. All six members of this component team sat at a table in proximity to each other and could freely communicate and interact during the simulation.

The Support Team (Intelligence). Each of the four lower-level members of this component team had the ability to generate queries about any one space from one specific source: human intelligence, allied intelligence, visual imagery, and communication intercepts. These different types of assets had different accuracy probabilities, and each had a 95 percent probability to be accurate in a predetermined area of the game grid (northern, southern, central, or border region). An important task of the multiteam system was to find where the intelligence assets were accurate; these areas were their sweet spot. This information had important performance implications for the multiteam system, because intelligence assets could deliver misleading information when they were sent to non-sweet spots. Each member of the intelligence team could generate eight queries per round from one of the four different sources; thus, the multiteam system as a whole could utilize a total of up to 32 intelligence assets per game round. The director of intelligence and assistant director of intelligence evaluated these missions and could assist their component team by making additional moves. All six team members sat at a table in proximity to each other and could freely communicate and interact during the simulation.

The Leadership Team. The Leadership Team consisted of six individuals that included both pairs of directors and assistant directors, in addition to the mission commander and vice-commander. As described earlier, the directors and assistant directors were boundary spanners who linked the leadership team to the two different component teams. The commander provided overall guidance for the multiteam system. The vice-commander was responsible for maintaining an up-to-date mental picture of the battle space on the COP, the central focal point for integrating information obtained from all three component teams that was supposed to represent the team's shared understanding of the task environment at any one moment. Within the overall leadership team, the

mission commander and directors were primarily responsible for overseeing future operations, whereas the vice commander and assistant directors were responsible for recording the results and findings of past operations. All four team members sat at a table in proximity to each other and could freely communicate and interact during the simulation, however, the directors and assistant directors moved back and forth from the Leadership Team to their component team as part of their boundary-spanning role.

The simulation required reciprocal interdependence, in the sense that missions conducted by component teams were the input for the activities of the Leadership Team (who could edit decisions), but the decisions of the Leadership Team also served as inputs for the component teams.



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